STAT301 Homework 8 Name CHARLES SANDAHL

1. Find covariance matrix of A=$\begin{pmatrix} 2 & 2 \\ 4 & 0 \end{pmatrix}$ by hand.

$$\begin{bmatrix} 2 & 2 \\ 4 & 0 \end{bmatrix} - \frac{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ 4 & 0 \end{bmatrix}}{2} = \frac{\begin{matrix} 6 & 2 \\ 6 & 2 \end{matrix}}{2} \quad \begin{bmatrix} 2 & 2 \\ 4 & 0 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$$

2. If a rock is thrown upward on the planet Mars and its height after t seconds is given by $H(t) = 10t - 1.86t^2$. Find the instantaneous velocity at $t = 2$ seconds. (Note: the instantaneous velocity is the derivative of the position function $H(t)$)

$10t - 1.86t^2$

$1.86 \cdot 2 \cdot t^{2-1}$

$10 \cdot 1t^{1-1}$ $\quad 3.72 \cdot t^1$

$10 \cdot 1$

$10 - 3.72t$

$10 - 3.72(2)$

$10 - 7.44$

2.56

3. Find the derivatives of the following functions:

a) $f(x) = 3x^2 - 4x + 1$

$f'(x) = 3 \cdot 2 \cdot x^{2-1} - 4 \cdot 1 \cdot x^{1-1} + 0$

$6x - 4$

b) $g(x) = x^{-2}$

$g'(x)=$

$-2 \cdot x^{-2-1}$

$-2x^{-3}$

c) $h(x) = \frac{2x+1}{x+3}$

$h'(x)= (2 \cdot 1 \cdot x^{1-1})(x+3) - (2x+1)(1 \cdot x^{1-1})$

$2(x+3) \quad -(2x+1)(1)$

$2x+6 - 2x - 1$

$\frac{5}{(x+3)^2} = \frac{5}{x^2+9}$

d) $g(x) = x^2(e^x)$

$g'(x)= (2 \cdot 1 \cdot x^{2-1})(e^x) + (x^2)(e^x)$

$2x(e_x) + x^2(e^x)$

$e^x(2x + x^2)$

$3x^{3-1} - 3 \cdot 1x^{1-1} + 0$ $\quad 3(x^2 - 1)$

4. If $f(x) = x^3 - 3x + 6$ $\quad 3x^2 - 3$ $\quad 3(x-1)(x+1)$

a) Find the intervals where the function is increasing and where the function is decreasing.

$(-\infty, -1)$ = increasing

$(-1, 1)$ = decreasing

$(1, \infty)$ = increasing

b) Find the local minimum and local maximum of the function, and specify the value of the local minimum and maximum.

local minimum = 1

local maximum = -1

5. A box with a square base and open top must have a volume of 32000 cm^3. What is the dimensions of the box that minimize the amount of material used.

Step 1: Set up the function for the optimization function

$A = L^2 + 4Lh$

Step 2: Set up the constrain and rewrite one variable using the other

$V = L^2 h$ $\quad 32000 = L^2 h$ $\quad \frac{32000}{L^2} = h$

$2 \cdot L^{2-1} - \frac{128000}{L^{1+1}}$

$2L - \frac{128000}{L^2}$

$A = L^2 + 4L\left(\frac{32000}{L^2}\right)$ $\quad \frac{128000\cancel{L}}{L^{\cancel{2}}}$ $\quad A = L^2 + \frac{128000}{L}$

$\frac{2L^3 - 128000}{2} = L^3 - 64000$

$\frac{32000}{40^2} = \frac{32000}{1600} = 20 = h$

$\sqrt[3]{L^3 = 64000}^{\,2} = 40 = L$

Step 3: Use the first derivative test to find the maximum or minimum

$L = 60$

100^2

$10000 - 1280$

$L = 10$

$10^2 - \frac{128000}{10}$

$100 - 12800$

$(-\infty, 40)$ = decreasing

$(40, \infty)$ = increasing

Local min = 40

local max = ∞

6. Find the dimensions of a rectangle with perimeter 100 m whose area is as large as possible.

Step 1: Set up the function for the optimization function

$A = L \cdot W$

Step 2: Set up the constrain and rewrite one variable using the other

$P = 2L + 2W$ $\quad \frac{100 = 2L + 2W}{2} = 50 = L + W$

$L = 50 - W$

$A = (50 - W) \cdot W$

$50W - W^2$

$\frac{50 - 2W}{2} = 25 - W = 0$ $\quad W = 25$

$L = 50 - 25$

$L = 25$

Step 3: Use the first derivative test to find the maximum or minimum

$(-\infty, 25)$ = increasing

$(25, \infty)$ = decreasing

25 = maximum

$-\infty$ = minimum